

Linear Algebra Lecture Notes

Rank and Elementary Matrices

1 Row Rank and Column Rank

Let $A \in M_{m \times n}(F)$ be a matrix over a field F .

Definition 1 (Row Space). *The **row space** of A is the subspace of F^n spanned by the rows of A .*

$$\text{Row}(A) = \text{span}\{R_1, \dots, R_m\}.$$

Definition 2 (Column Space). *The **column space** of A is the subspace of F^m spanned by the columns of A .*

$$\text{Col}(A) = \text{span}\{C_1, \dots, C_n\}.$$

Definition 3 (Rank). *The **row rank** of A is*

$$\text{rk}_r(A) = \dim(\text{Row}(A)).$$

*The **column rank** of A is*

$$\text{rk}_c(A) = \dim(\text{Col}(A)).$$

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2 Row Rank Equals Column Rank

Theorem 1. *For every matrix $A \in M_{m \times n}(F)$,*

$$\text{rk}_r(A) = \text{rk}_c(A).$$

Proof. Consider the linear map

$$L_A : F^n \rightarrow F^m, \quad x \mapsto Ax.$$

The image of this map is exactly the column space of A :

$$\text{Im}(L_A) = \text{Col}(A).$$

Hence

$$\text{rk}_c(A) = \dim(\text{Im}(L_A)).$$

By the rank-nullity theorem,

$$\dim(\ker L_A) + \dim(\operatorname{Im}(L_A)) = n.$$

Therefore

$$\operatorname{rk}_c(A) = n - \dim(\ker L_A).$$

Now perform row operations on A to obtain a row-equivalent matrix A' .
Row operations do not change the solution set of

$$Ax = 0.$$

Hence

$$\ker(L_A) = \ker(L_{A'}).$$

Thus

$$\operatorname{rk}_c(A) = n - \dim(\ker L_{A'})$$

But in row echelon form, the number of nonzero rows equals the dimension of the row space.
Therefore

$$\operatorname{rk}_r(A') = \text{number of nonzero rows of } A'.$$

These rows form a basis of the row space, so

$$\operatorname{rk}_r(A') = n - \dim(\ker L_{A'}).$$

Thus

$$\operatorname{rk}_r(A') = \operatorname{rk}_c(A).$$

Since row operations preserve row rank,

$$\operatorname{rk}_r(A) = \operatorname{rk}_r(A').$$

Therefore

$$\operatorname{rk}_r(A) = \operatorname{rk}_c(A).$$

□

Definition 4 (Rank). *Because row rank and column rank coincide, we define*

$$\operatorname{rk}(A) = \operatorname{rk}_r(A) = \operatorname{rk}_c(A).$$

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3 Elementary Matrices

Definition 5. *An **elementary matrix** is a square matrix obtained from the identity matrix by performing a single elementary row operation.*

There are three types.

3.1 Type I: Row Scaling

Multiply a row by a nonzero scalar.

$$E = \text{diag}(1, \dots, 1, \alpha, 1, \dots, 1), \quad \alpha \in F^\times$$

3.2 Type II: Row Swap

Swap two rows.

$$E = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & 0 & 1 & \\ & & 1 & 0 & \\ & & & & \ddots \end{pmatrix}$$

3.3 Type III: Row Addition

Add a multiple of one row to another.

$$E_{ij}(\alpha)$$

which has

$$a_{ij} = \alpha$$

and all other entries identical to the identity matrix.

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4 Row Operations as Matrix Multiplication

Proposition 1. *Performing a row operation on a matrix A is equivalent to left multiplication by an elementary matrix.*

$$A' = EA$$

Proof. Let E be the elementary matrix obtained by performing the row operation on the identity matrix.

Multiplying A by E applies the same operation to the rows of A .

Thus

$$A' = EA$$

is exactly the matrix obtained from the row operation.

□

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5 Invertibility of Elementary Matrices

Proposition 2. *Every elementary matrix is invertible.*

Proof. Each elementary row operation can be reversed:

- scaling by α is reversed by scaling by α^{-1}
- swapping rows is its own inverse
- adding αR_j to R_i is reversed by adding $-\alpha R_j$

Thus every elementary matrix has an inverse which is also elementary.

□

6 Matrix Inversion via Row Reduction

Let A be an $n \times n$ matrix.

To compute A^{-1} we solve

$$AX = I_n.$$

This can be done by row reducing the augmented matrix

$$[A \mid I_n].$$

If

$$EA = I_n$$

then

$$A^{-1} = E.$$

Equivalently

$$[A \mid I] \rightarrow [I \mid A^{-1}].$$

Remark 1. *If A cannot be reduced to the identity matrix, then A is not invertible.*